

SAN RAFAEL, Argentina

WMO: 875090

Lat: 34.5908S

Lon: 68.4017W

Elev: 2454

StdP: 13.44

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.5	31.1	9.6	9.8	51.5	14.1	12.3	48.1	23.1	57.2	19.4	56.5	3.5	270	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.6	94.8	67.1	92.0	66.4	89.4	65.4	71.5	88.2	69.7	86.0	68.1	83.6	11.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	106.8	78.7	64.5	99.5	77.1	62.6	93.1	75.7	37.0	88.1	35.3	86.3	33.9	83.9	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	16.9	14.1	23.8	100.6	2.4	3.0	22.1	102.7	20.7	104.4	19.3	106.1	17.6	108.2		
(5)			21.8	75.3	2.1	2.4	20.3	77.0	19.1	78.4	17.9	79.8	16.4	81.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	74.9	72.2	67.6	59.9	53.3	48.1	46.8	50.2	54.8	60.8	67.0	72.0		
	DBStd	11.41	5.21	6.22	6.39	6.39	6.11	5.51	6.16	7.10	6.84	6.52	6.02	5.81		
	HDD50	395	0	0	1	6	35	98	134	83	31	6	1	0		
	HDD65	2668	3	15	48	175	364	507	564	460	310	162	48	12		
	CDD50	4256	770	623	545	303	138	41	36	90	176	339	512	683		
	CDD65	1052	308	217	127	22	2	0	0	2	5	30	109	230		
	CDH74	12442	3463	2307	1351	322	40	2	6	77	169	545	1422	2738		
	CDH80	5048	1622	1003	478	60	5	0	0	16	30	142	510	1182		
(14) Wind	WSAvg	5.6	6.0	5.7	5.3	4.6	4.5	4.9	5.3	5.8	5.9	6.3	6.6	6.6		
(15) Precipitation	PrecAvg	13.2	2.1	1.9	1.7	0.9	0.4	0.4	0.4	0.6	0.9	1.2	1.6	1.8		
	PrecMax	23.5	8.0	11.1	7.0	3.6	2.1	3.1	2.0	3.1	4.4	4.2	6.3	6.6		
	PrecMin	4.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	4.7	1.8	1.8	1.5	1.0	0.5	0.6	0.5	0.7	1.0	1.1	1.3	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	99.3	97.1	92.4	85.5	78.7	72.6	73.2	81.9	84.0	88.8	93.4	97.5		
		MCWB	69.2	68.8	67.3	63.0	58.4	53.0	51.0	56.3	57.2	61.8	64.4	66.6		
	2%	DB	95.1	93.1	88.6	81.5	73.6	67.7	68.2	74.9	78.7	84.0	89.5	93.6		
		MCWB	67.8	67.3	65.9	61.6	57.1	50.5	49.6	53.4	55.6	60.0	63.5	65.7		
	5%	DB	91.8	89.7	85.5	77.6	69.8	64.1	64.2	69.7	74.7	80.0	86.0	90.0		
		MCWB	66.8	66.1	65.1	59.6	55.1	49.2	48.1	51.1	54.2	58.2	62.1	65.3		
	10%	DB	88.7	86.4	82.1	73.9	65.9	60.6	60.1	64.9	70.2	76.4	82.3	86.7		
		MCWB	65.9	65.4	63.5	58.2	53.3	47.5	46.2	48.5	52.2	56.8	60.4	63.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.9	73.7	71.9	66.7	61.2	55.1	53.8	58.0	61.1	66.4	69.4	73.4		
		MCDB	91.1	89.4	86.8	78.3	74.4	67.8	68.4	77.4	78.8	82.5	86.6	89.7		
	2%	WB	71.1	71.3	68.9	63.8	58.7	52.3	51.0	54.4	57.7	63.0	66.3	69.7		
		MCDB	88.6	87.3	82.9	76.6	69.9	64.5	65.6	72.8	74.4	78.5	84.0	86.8		
	5%	WB	69.4	69.2	66.6	61.6	56.4	50.2	48.9	51.8	55.3	60.5	63.9	67.3		
		MCDB	86.8	83.7	81.3	74.0	66.4	61.5	62.2	67.3	71.4	75.6	81.2	84.0		
	10%	WB	67.8	67.4	64.8	59.6	54.3	48.3	46.8	49.4	53.0	58.2	61.9	65.5		
		MCDB	84.2	81.3	79.2	71.1	63.4	59.3	58.3	63.2	68.0	72.4	78.6	82.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.6	24.3	23.0	22.1	20.9	22.6	23.3	24.2	24.5	24.5	26.3	26.1		
		MCDBR	30.3	29.6	29.0	28.8	27.3	28.3	29.6	31.9	32.2	31.7	32.5	31.3		
		MCWBR	11.6	11.2	11.9	13.8	14.2	15.8	16.0	15.8	15.4	14.3	14.2	12.8		
		MCDBR	26.5	24.8	25.2	24.9	23.8	26.1	27.2	29.0	28.9	27.5	28.5	27.0		
	5% WB	MCWBR	11.3	11.1	11.7	13.1	13.4	15.4	15.6	15.2	14.9	14.0	14.2	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.386	0.374	0.363	0.349	0.324	0.297	0.296	0.320	0.356	0.354	0.361	0.372		
	taud		2.361	2.400	2.426	2.434	2.467	2.546	2.517	2.438	2.338	2.374	2.370	2.373		
	Ebn at Noon		302	298	289	275	265	267	273	281	287	301	307	307		
	Edh at Noon		42	38	35	31	26	23	24	30	37	39	41	41		
(44) All-Sky Solar Radiation	RadAvg		2543	2217	1810	1341	938	838	915	1200	1625	2033	2463	2637		
	RadStd		139	120	132	128	108	69	65	88	109	199	144	132		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.53	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A	N/A	+113

Nomenclature: See separate page